EARTH

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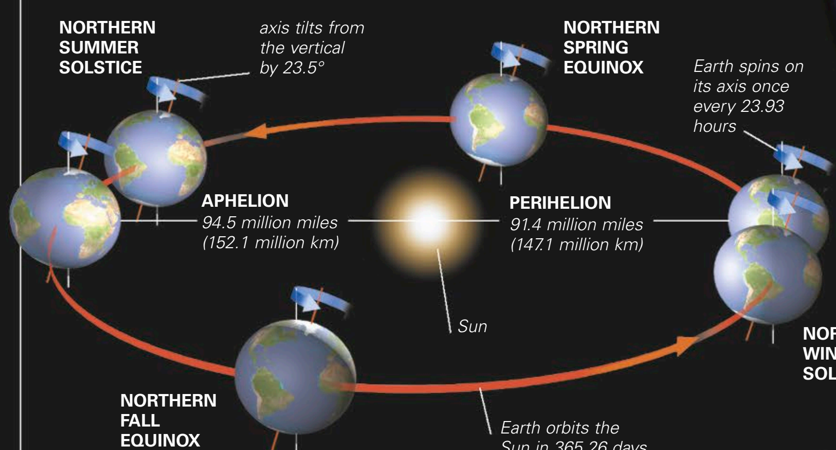
EARTH IS THE THIRD-CLOSEST PLANET to the Sun. The largest of the four rocky planets, it formed approximately 4.56 billion years ago. Earth's internal structure is similar to that of its planetary neighbors, but it is unique in the solar system in that it has abundant liquid water at its surface, an oxygen-rich atmosphere, and is known to support life. Earth's surface is in a state of constant dynamic change as a result of processes occurring within its interior and in its oceans and atmosphere.

ORBIT

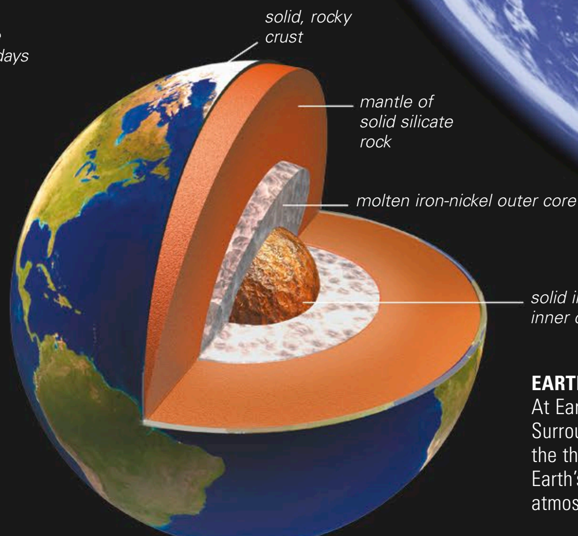
Earth orbits the Sun at an average speed of 67,000 mph (108,000 kph) in a counterclockwise direction when viewed from above its North Pole. Like the other planets, Earth orbits the Sun along an elliptical path. As a result, about 7 percent more solar radiation currently reaches Earth's surface in January than in July. The plane of Earth's orbit around the Sun is called the ecliptic plane. Earth's spin axis is not perpendicular to this plane but is tilted at an angle of 23.5°. The eccentricity of Earth's elliptical orbit around the Sun (the degree to which it varies from a perfect circle) changes over a cycle of about 100,000 years, and its axial tilt varies over a 42,000-year cycle. Combined with a third cycle—a wobble in the direction in which the spin axis points in space, called precession (see p.64)—these variations are believed to play a part in causing long-term cycles in Earth's climate, such as ice ages.

SPIN AND ORBIT
Earth is about 3 percent closer to the Sun at perihelion (in January) than at aphelion (in July). Its axial tilt combined with its elliptical orbit gives rise to seasons (see p.65).



STRUCTURE

Earth's rotation causes its equatorial regions to bulge out slightly, by about 13 miles (21 km) compared to the poles. Internally, Earth has three main layers. The central core has a diameter of about 4,350 miles (7,000 km) and is made mainly of iron with a small amount of nickel. It has a central solid part, which has a temperature of about 8,500°F (4,700°C), and an outer liquid part. Surrounding the core is the mantle, which contains rocks rich in magnesium and iron and is about 1,700 miles (2,800 km) deep. Earth's crust consists of many different types of rocks and minerals, predominantly silicates, and is differentiated into continental crust and a thinner oceanic crust.



EARTH INTERIOR
At Earth's center is a hot, dense core. Surrounding the core are the mantle and the thin, rocky outer crust that supports Earth's biosphere, with its oceans, atmosphere, plants, and animals.